Project Analysis

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The purpose of the analysis is to summarize the project. Each of the devices that were configured will be given a summary of how they were configured.

Router 1 (R1) was configured to connect to PC1, PC2, Kali Linux and Switch 1 (S1) on a separate subnet. The subnet for these devices is 192.168.10.64/29. R1 was configured and worked as designed. Interface Ethernet 0/1 was connected to R2. R1 was given a sub interface, Ethernet 0/0.10, to connect to S1 with the address 192.168.10.65. OSPF routing protocol was configured to connect to the R3 subnet.

S1 was configured to allow PC1, PC2, and Kali Linux to access VLAN10. VLAN10 was given the address 192.168.10.66. Port-security was enabled on all switch ports with access to VLAN10. Interfaces Ethernet 0/0-2 was connected to each of the end devices. Trunking was set on interface Ethernet 0/3 to trunk VLAN10 to R1.

Router 2 (R2) was configured to connect R1 to Router 3 (R3). OSPF routing protocol was configured on R2 to connect the routers. Ethernet 0/0 was configured with the address of 192.168.20.2. Ethernet 0/1 was configured with the address of 192.168.30.1. There is a default static route from R2 to the internet via 192.168.83.2 and attached to Ethernet 0/2. R2 included an extended access-list (ACL), access-list 100, to allow the subnets of R1, R2, and R3 to connect to the internet.

R3 is on a different subnet than R1. The subnet for R3 is 192.168.40.0/24. Interface Ethernet 0/0 is connected to R2 with an address of 192.168.30.2. OSPF routing protocol was configured on R3 to connect to R1.

PC1 was configured to connect to VLAN10 via S1. PC1 was given the address of 192.168.10.67/29.

PC2 was configured to connect to VLAN10 via S1. PC2 was given the address of 192.168.10.68/29.

The Kali Linux machine was configured to connect to VLAN10 via S1. The Kali Linux machine was given the address of 192.168.10.69/29. John the Ripper, Nmap, and Metasploit were tools used for penetration (pen) testing. The Nmap tool was used to scan the subnet of R3. Nmap returned ports 21, 22, and 80 were open on host 192.168.40.100; which is the address of the Ubuntu server. The protocols for the port numbers are as follows: File Transfer Protocol (FTP), Secure Shell (SSH), and Hypertext Transfer Protocol (HTTP). The Metasploit tool was used to sniff passwords on the command line. The password sniffer was psnuffle which is in the auxiliary/sniffer/psnuffle folder in MSF console. The password and username for the Ubuntu server was found by logging in via FTP. To retrieve the password file, SSH login was used. The password was entered to log into the machine. Since the user of the Ubuntu server is a sudo user, escalating privileges was used by typing sudo SU root. From there, locating the /etc/shadow file was easy. The cp command was used to copy the file to the home directory of the user (/home/user). The file was given a different name. The permissions were changed using chmod and chown commands to allow for reading of the file without being the root user. Exit the SSH connection to return to the Kali Linux machine. Now, log in to the ftp server and download the shadow file using the get command. Next, use John the Ripper to crack the password file. The password for the root user will be shown in plaintext.

The Ubuntu Server was configured on the same subnet as R3. Ubuntu Server was given the address 192.168.40.100/24. The server would run the apache2 service to create a web page for the company network to display information. The Ubuntu server was also configured to run an SSH and FTP server for the network.